CSB - Lec 21

Enrichment Analysis

Goal: Identifying functional patterns in (gene) expression data.

Define the experimental design → Collect the biological samples → Generate the expression data → Identify the Differential Genes → Identify the Functional Groups

Data → Relation? Biological functions
↳ Characterized by gene expression?

Identification of Functional Groups

Gene sets: Score the set depending on the gene expression of its member genes.

Networks: Just visual, or Identify modules satisfying some joint gene expression and topology requirement

Pathways: Just visual, or Score the pathways exploiting gene expression and topology

What is gene-set enrichment analysis?

→ with gene expression data

Break down cellular function into gene sets.

↳ Every set of genes is associated to a specific cellular function, process, component or pathway

GSEA: method to identify classes of genes that are over-represented in a large set of genes and may have an association with different phenotypes;
method that determines whether an a priori defined set of genes shows statistically significant, differences between two biological states

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GESA

→ Gene expression data can be related to gene sets in order to mine its functional meaning

→ Which gene-sets summarize at best gene expression pattern?

↳ differ between: up regulated, down regulated, not significant
↳ meaning of significant enrichment

Gene Functions

→ refers to the specific roles performed by individual genes/sets

Gene Functions → Gene Ontology

→ GO is a source on the information of gene functions

→ structured controlled vocabulary

→ GO: a collection of terms describing gene products and their definitions and the logical relationships between them. eg: numbers

↳ represented as DAG: relations: „part of" „is a"

↑ relationships are not symmetrical

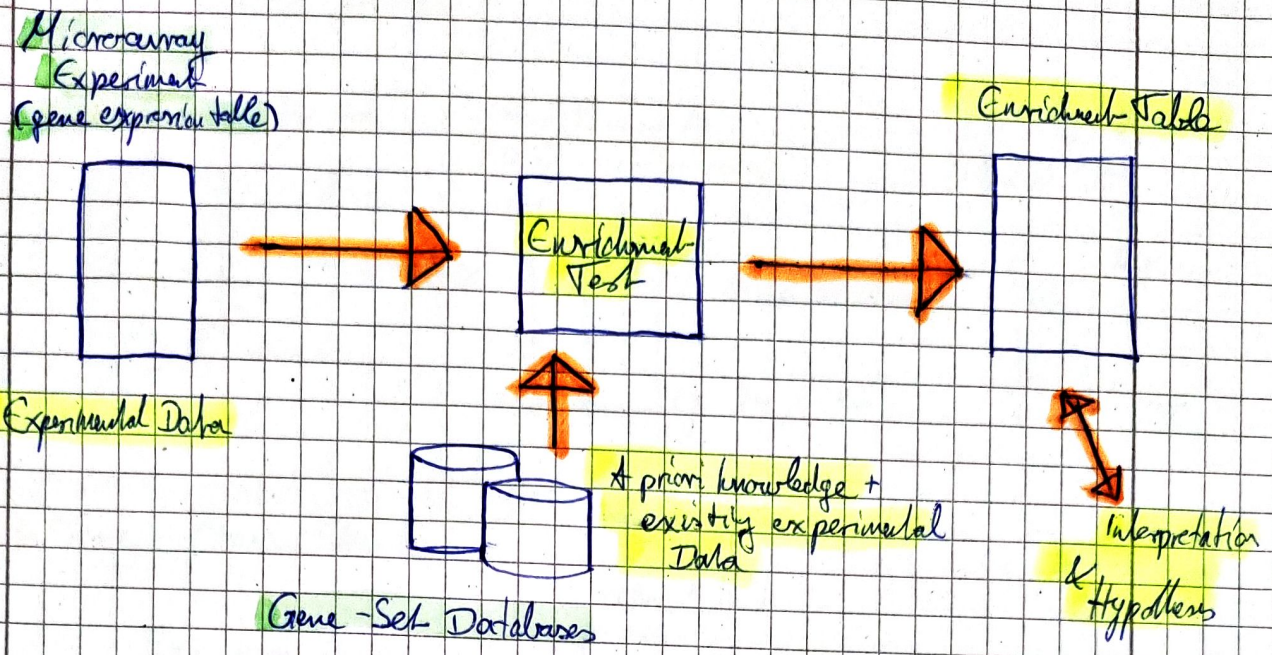
→ Consists of three different ontologies:

- cellular component
- molecular function
- biological process

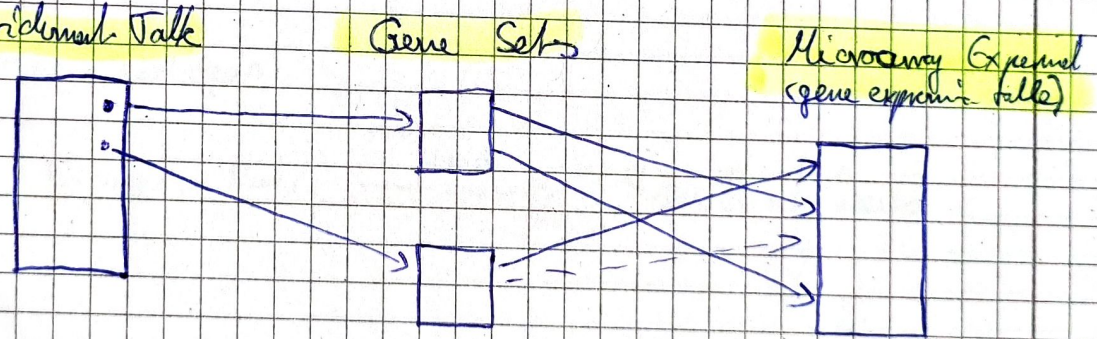
GSE

Gene Set Enrichment Analysis

Enrichment Test -> Statistical Test



Statistical Test used to determines whether a particular set of genes is enriched within a given dataset compared to what would be expected by chance.



Maps: Enrichment Table -> Gene Sets -> Gene Expression Table

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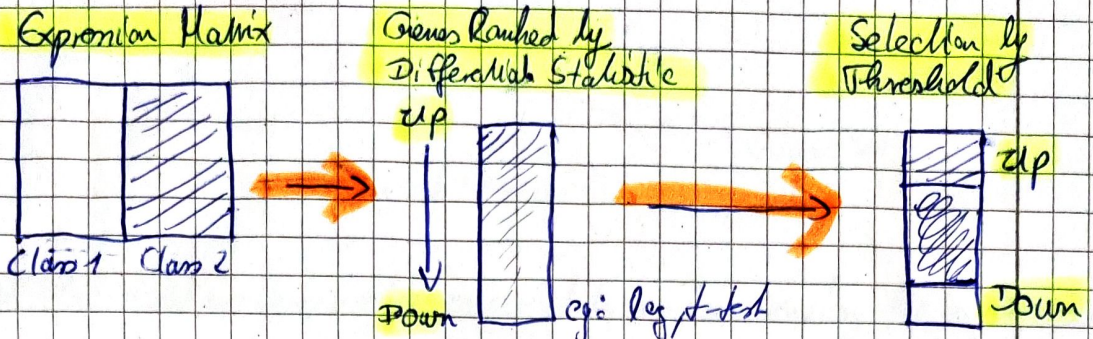
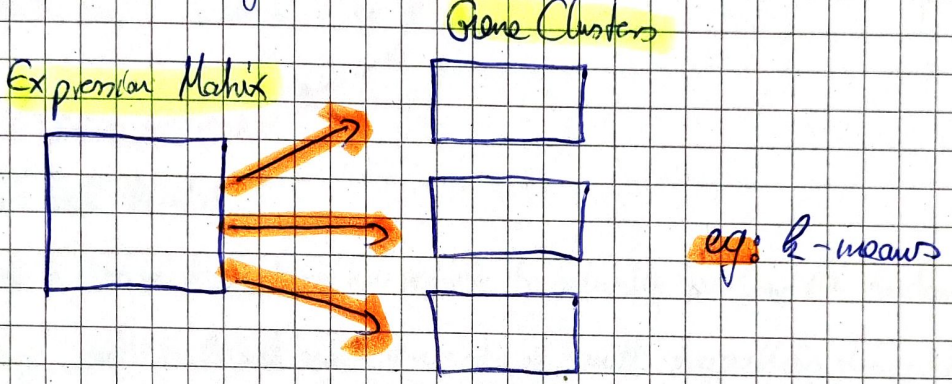
Enrichment Test

How to Test?

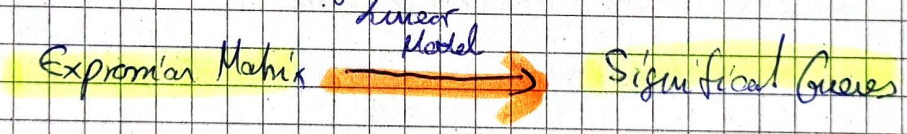
- **Two-class Design:**



- **Time-Course Design:**



- **Other Designs**



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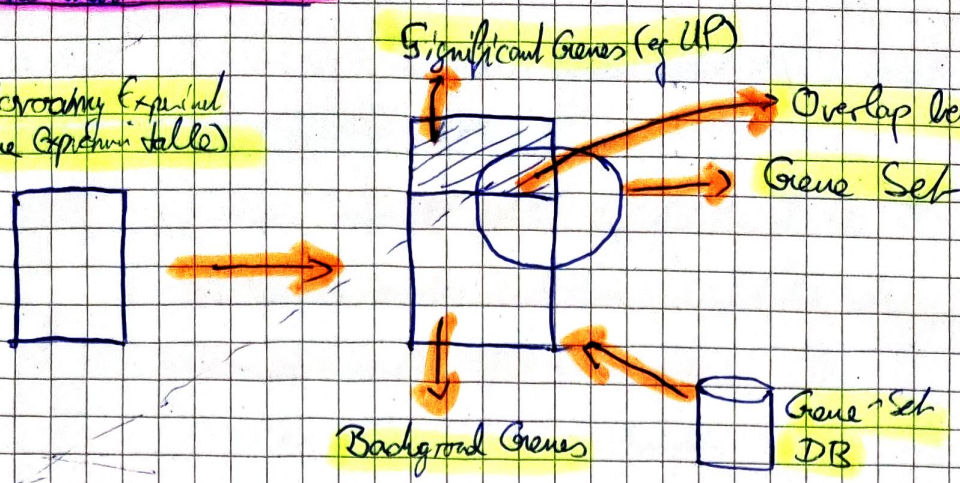
Enrichment Test



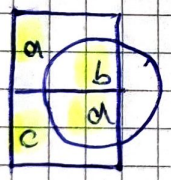
→ Is this overlap larger than expected by random sampling the array genes?

↳ to check this take random samples from gene expression table and check. or

Fishers Exact Test

→ Statistical Model

→ Fisher's exact test does not require to actually perform the random sampling, it is based on a theoretical null-hypothesis distribution



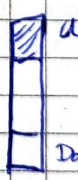
→ Formula (a, b, c, d, n) → Enrichment p-value

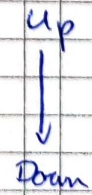
$$= \frac{(a+b)!\,(a+c)!\,(c+d)!\,(b+d)!}{n!\,a!\,b!\,c!\,d!}$$

Beyond Fishers Test

Enrichment Test

- Threshold-dependent
  eg: Fishers test
- Whole Distribution
  eg: GSEA





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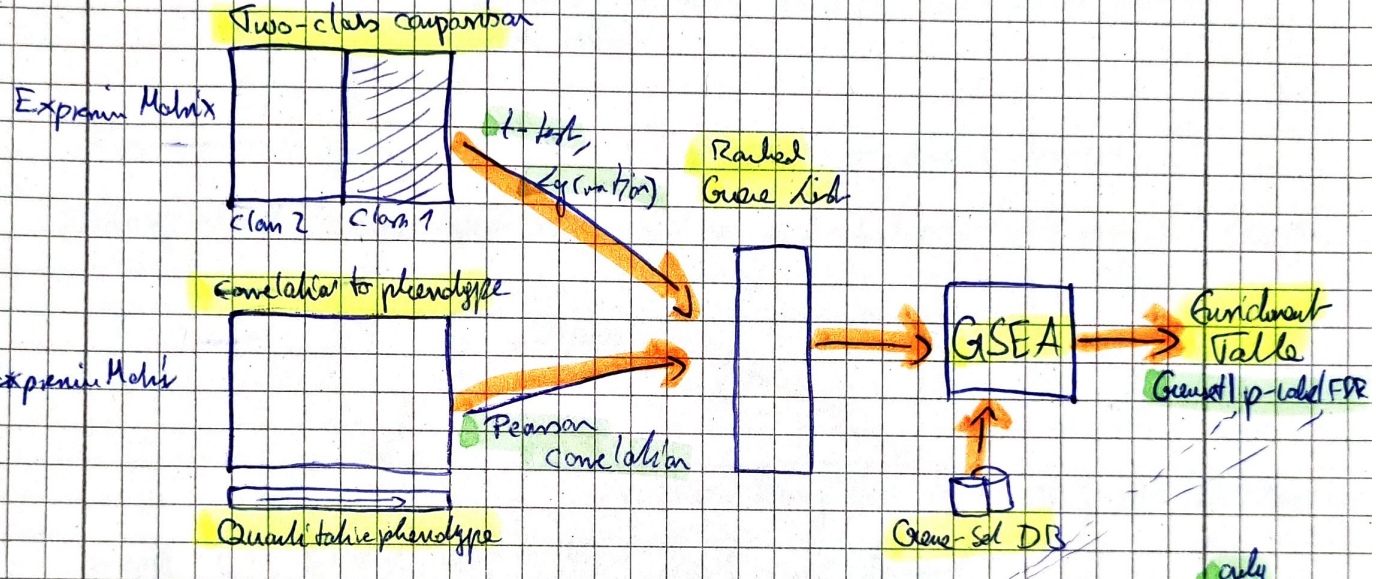
Beyond Fisher's Test

- Rank-based methods have been shown to be more stable and statistically powerful
  - ↳ not "natural" value for the threshold ↯
  - ↳ Different results at different threshold settings ↯
  - ↳ Loss of information due to threshold ↯
    - ↳ no resolution between strengths

=> Use rank-distribution whenever possible

GSEA Enrichment Test



only

↳ p-value depends on the single gene set performance

→ FDR depends on the performance of all gene-sets

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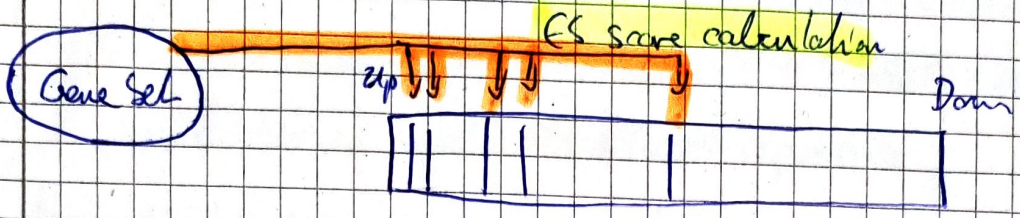
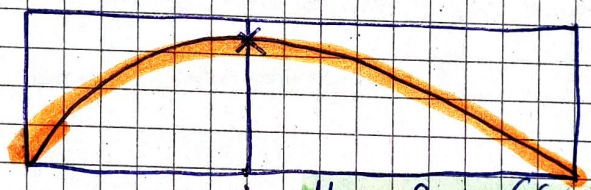
GSEA: Method

Steps:
1) Calculate the ES score
2) Generate the ES distribution for the null hypothesis using permutations
3) Calculate the empirical p-value
4) Calculate the FDR



Is the distribution random or is there an enrichment in either end?

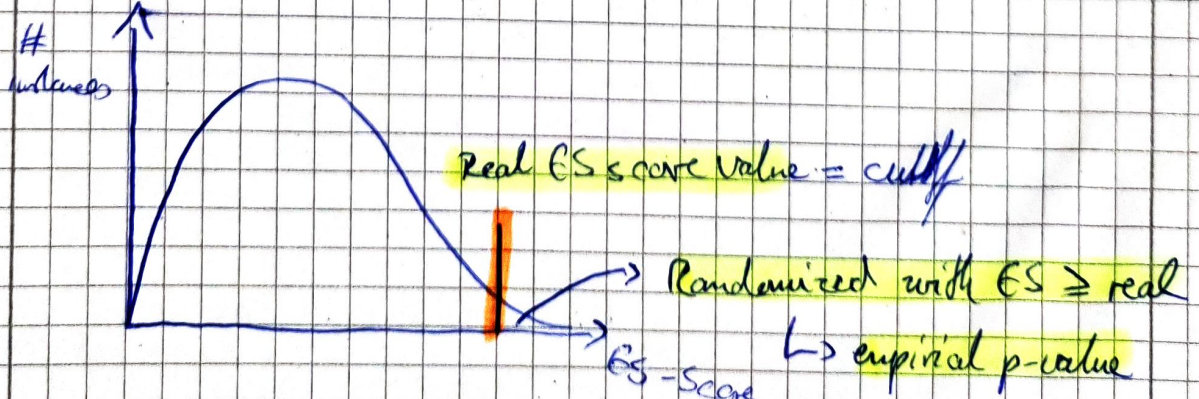
Every present gene (vertical bars) gives a positive contribution,
Every absent gene gives a negative contribution to the running ES score



↳ Max Running ES score → final ES score

High ES score ↔ High local enrichment

→ Empirical p-value estimation (∀ gene sets)
1) Generate null hypothesis distribution from randomized data
2) Estimate empirical p-value



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De novo network enrichment (KeyPathwayMiner)

Look at the graphs not at the genes anymore and identify network enrichments.

Individual Nodes Exceptions (INES) model

In: Interaction Networks

Goal: Find $\forall$ maximal connected subnetworks where all but at most K nodes/genes are differentially expressed in all but at most L samples/patients

INES: L for each node/gene individually

1) Apply the statistics of choice to the expression data
2) Find $\forall$ genes that are not 'active' in all cases by at most L
3) Extract all maximal connected components, where all nodes but at most K (gene exceptions) are 'active' in all but at most L cases (case exceptions)

$L = x$, check if a node has less than or equal to x number of 0's
↳ then node is active!

② Groups where ~~only~~ $\leq K$ nodes are non-active

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Global Node Exceptions (GLONE) model

In: Interaction Networks

Goal: Find ∀ maximal connected subnetworks where all nodes/genes are differentially expressed in all but at most k samples/patients

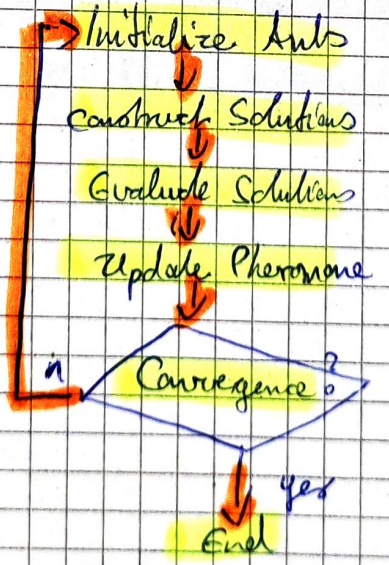
GLONE: k parameter is global

For every node count zeros! → Sum $\leq k$

Ant Colony Optimization

Summary:

- A set S of ants are randomly placed on the working space in graph G
- Each ant tries to construct a valid solution to the problem by random walk
- The probability that an ant stady on vertex u moves to vertex v
- Once an ant s has successfully constructed a valid solution to the problem, it can update the pheromone value of the edges if visited



Eg: Travelling Salesman Problem

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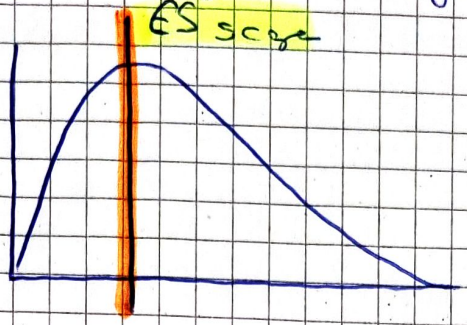
Enrichment Analysis

(1) Gene Set Enrichment Analysis

a) For the statistical assessment in the enrichment analysis, we need to order them by decreasing enrichment score

b)



c) Way to calculate the significance of the overlap between a set of significant genes and a pre-given set of genes:
Statistical test or strategy: Fishers Exact Test, (GSEA Enrichment Test)

(2) Gene Ontology

Gene Ontology is a controlled vocabulary that defines biological functions and entities and relationships between them.
Typically examples of the latter are "is a" and "part of".
The GO is a free, IT term or directed graph that is acyclic

Enrichment Analysis

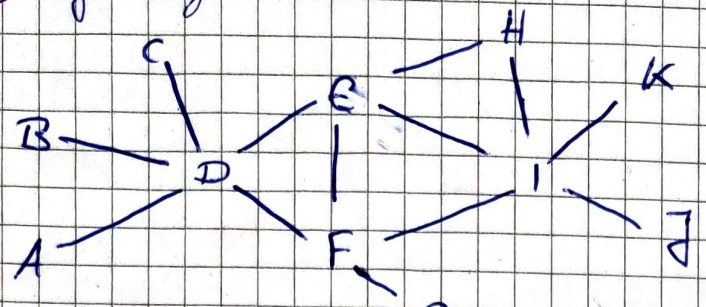
(3) Ant Colony Optimization

a) Traveling Salesman Problem:

Finding the shortest possible route that visits a set of cities and returns to the starting city, while visiting each city once.

Candidate Analysis

(4) KeyPathway Miner



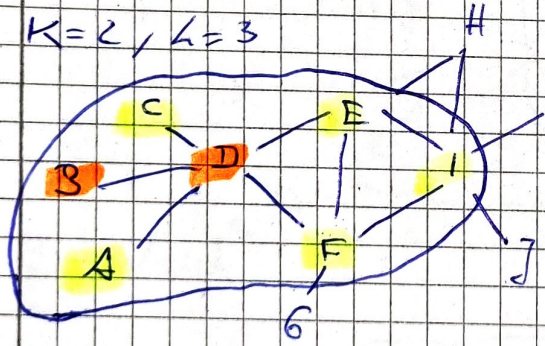
✓ : active
✗ : not active

active if less or equal ✓ zeros (not counting ...)

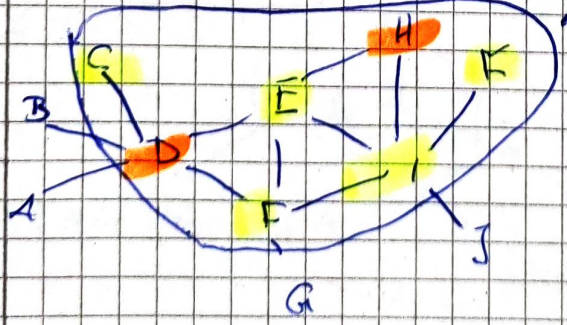
	P1	P2	P3	P4	P5	L=1	L=3
A	1	1	1	0	0	✗	✓
B	0	0	0	0	1	✗	✗
C	0	1	1	1	1	✓	✗
D	0	0	0	1	0	~~✗~~	~~✗~~
E	1	1	0	0	1	~~✗~~	✓
F	1	1	1	1	1	✓	✓
G	1	0	0	0	0	✗	✗
H	0	0	1	0	0	✗	✗
I	1	1	1	1	1	✓	✓
J	0	0	0	1	1	✗	✓
K	0	1	0	0	1	✗	✓

(4) Else INES

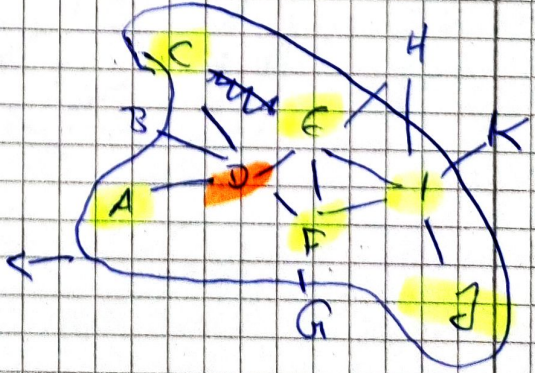
K=2, L=3



→ Not valid since K & J could be also in it



→ Not valid since A & K & J could be also in it



Not valid since K is not in it

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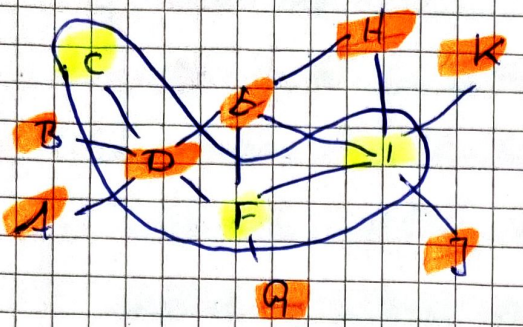
Equivalent Analysis

(9)

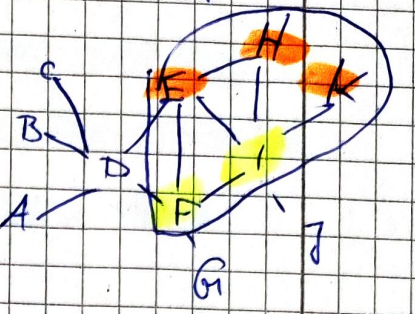
a) INES

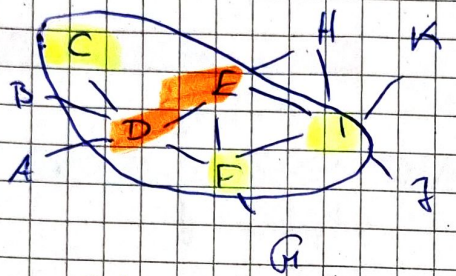
K = 1 , L = 1



→ Valid

Not valid

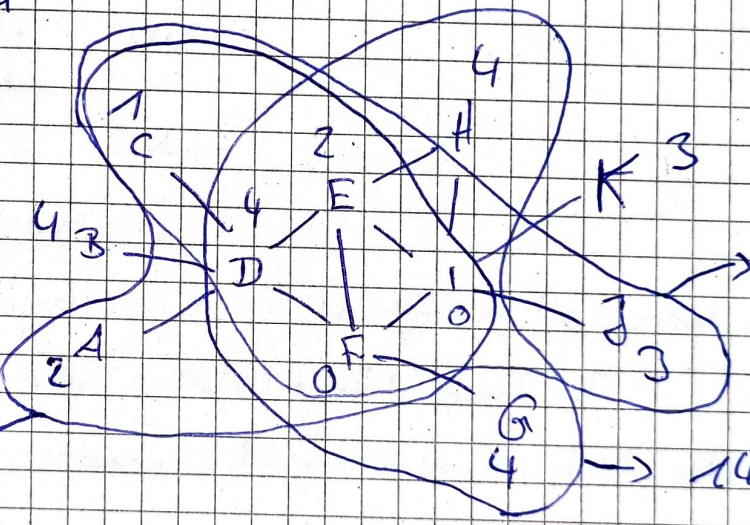




→ Not Valid

b) GLOVE

L = 10

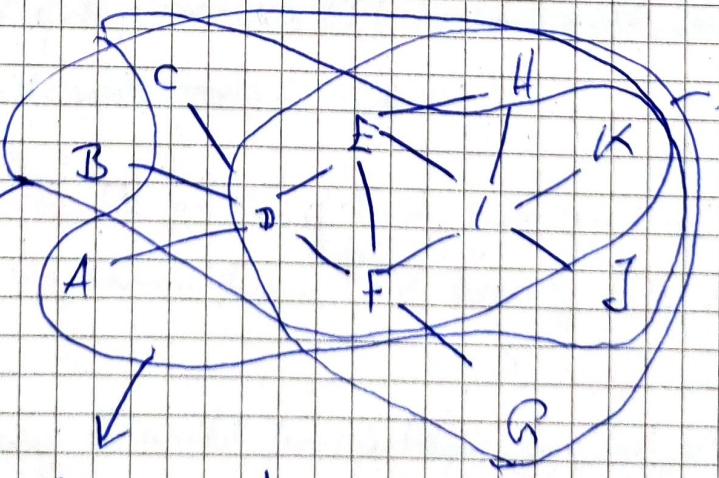


10 → Valid

9 → Valid

14 → Not Valid

L = 15



20 → Not Valid

18 → Not Valid

13 → Valid

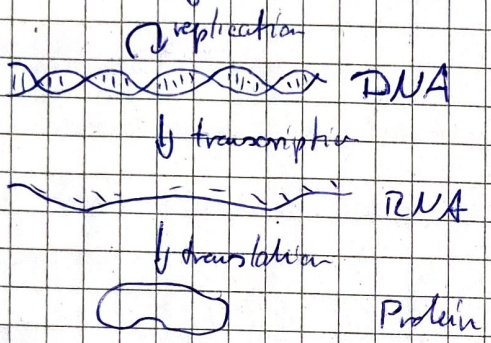
CSB - 02

Sequence logos, Sequence Motif Discovery (document)

- **Sequence motifs:**
  - → short, conserved sequences that are shared among a group of related sequences
  - → are typically associated with a particular biological function
- **Motif Discovery:** → Identification of Transcription Factor Binding Sites
  - → a procedure of finding enriched sets of similar short sequences in a large sequence dataset

→ gene regulation, protein function, evolutionary analysis / expression protein motif

Central Dogma



Transcription Factor

→ Knock out a specific TF ⇒ See differentially expressed genes
→ That are likely to be controlled by that specific TF
→ Their upstream sequences are likely to contain the binding sites

Given: A set of long sequences likely to contain a short, conserved sequence motif
Task: Find the most prominent motif

Problems: → the position of binding sites can vary ↯
→ Common feature (binding sites) are typically very short ↯

→ Motif Discovery is useful for identifying enriched sets of similar short sequences in a large sequence dataset, such as identifying transcription factor binding sites or protein motifs

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How to represent a motif?

→ Regular expressions :

I) Consensus eg: AAGAAA

II) Degenerate sequence A[A/T]BW

→ Positional Count Matrix

→ Positional Probability Matrix

→ Positional Weight Matrix : $S = \log_2 (P_{ij} / \pi_j)$

probability of each letter at each position

background frequency

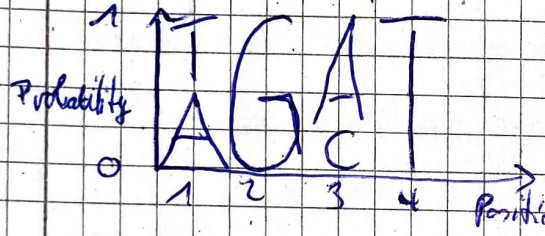
↳ $P_{pseudo}(N) = \frac{c_N + P_n}{\Sigma c + p}$ // use pseudocount to avoid 0

→ Sequence Logos

Sequence Logos

→ Shows which positions are more important than others

→ Any sequence of DNA, RNA or Protein can be represented as sequence logo



Information :

→ general consensus of the sequences

→ letter representing nucleotides / amino acids

→ (Ats.) are stacked on top of each other

→ order of predominance of the residues at every position

→ relative frequencies of every residue at every position

→ Amount of info present at every position in the seq measured in bits

→ Initiation point, cut point or other significant location

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Sequence Logos

- Profile: P_{ij}, $\forall i \in [1, L]$, $\forall j \in [1..|AAs|]$
  a probability distribution of AA's $\forall$ position, L = length of seq
- The uncertainty / entropy of distribution P_i at the i-th position of the profile:

$$H(P_i) = -\sum_{j \in AAs} P_{ij} \log_2 P_{ij}$$

$H(P_i) = 0$ => only one residue is found at that position
$H(P_i)$ is max => all residues have an equal frequency.

- Information Content of position i:

$$I(P_i) = \log_2 |AAs| - H(P_i)$$

- Background frequencies of AA's
  - ↳ Count AA occurrences in all known proteins, or only in the proteins of the superfamily under consideration
- Relative Entropy

$$H(P_i \| \pi) = -\sum_{j \in AAs} P_{ij} \log_2 (P_{ij} / \pi_j)$$

(π_j ↳ background frequency of AA$_j$)

- contribution of a residue: $P_{ij} \cdot H(P_i \| \pi)$
- heights of residues in a logo are proportional to their contributions
- stack height is proportional to the relative entropy
- colors highlight different properties

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How to find a motif?

→ eg: wet-lab experiments
  ↳ not exact enough
  ↳ ~~Mutation~~ Experiments to verify Binding sites are expensive & time consuming
    (Mutation of the BS?, Functional changes)
  ↳ Bad resolution
→ known are approximate positions
→ assume that BS are evolutionary conserved
→ Issues: → motifs can be different
  → Very short sequences in large DNA
    ↳ Bad Signal (FG) to noise (BG) ratio

Def

→ Motif: A recurring nucleotide sequence a (ℓ, d)-Motif, if the sequence is of length ℓ and has at most d mutations

→ Planted Motif Problem:

Given: A set S of sequences of length n and two ints ℓ, d with $d < \ell < n$

Wanted: A pattern M, such in each Sequence there is a substring which can be transformed to M with at most d substitutions

Variations:

find exactly one / several motifs
with / without gaps
in every sequence, ..

CGB

Find Motifs

→ Different Methods:
- → Bruteforce → $\Theta(4^l m n d)$
- → Statistical Approaches
- → Combinatorial Approaches

↳ use edit distance and brutforce all possible Motifs

Expectation Maximization

EM algorithm enables parameter estimation in probabilistic models with incomplete data.

e.g.: Coin Flip

Given: 2 Coins, A and B, biased (Θ_A, Θ_B)

Θ_A, Θ_B = likelihoods of seeing "Head"

X rounds of coin flips and Y flips per round

- Complete Information, MLE:
$$\frac{\#\text{Heads with Coin X}}{\text{Total Coins of Flip Coin X}} = \hat{\Theta}_X$$
- Incomplete Information, MLE: we don't know which coin produces which outcome
  - ↳ Iterative process:
    - → Guess initial Θ_A^0 / Θ_B^0
    - → Calculate which coin is the more likely one for what round
    - → use that for Θ_A^1 / Θ_B^1 using MLE
    - → Repeat until convergence
- This general Idea, EM: use probabilities for data completion & use MLE.
  - ↳ Now, every coin "votes" the series according to the likelihoods

  e.g.: Coin A votes 45% of series 1, thus 45% · 5T and 45% · 5H

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How to use EM for Motif discovery

→ Assumptions: → Motif, fixed length W
→ Motif is generated by a probability function
→ Background is generated by a different probability function

→ Unknown Parameters: → the motif itself, find different parameters of the probability function
→ the position of the Motif with the seq's

→ Algorithm:
→ Start with an initial guess of the motif positions Z^0
→ Calculate the first Motif representation Θ^0
→ use motif model to update the position likelihoods Z^{t+1} (E-step)
→ use these likelihoods to update the motif model Θ^{t+1} (M-step)
→ Iterate until convergence

→ Representation of a Motif:
→ represented by a probability matrix $\Theta = p_{c,k}$, with $p_{c,k}$ the probability of character c at position k
→ Same for BG

→ Likelihood of a starting position
→ represented by a matrix $Z = z_{i,j}$ with $z_{i,j}$ being the probability that the motif (given by Θ) starts at position j in sequence i

$P(X_i \mid Z_{ij}=1, \Theta)$: probability to find the motif in sequence X_i at position j for the given Motif model Θ

$$= \prod_{q=1}^{j-1} p_{c_i,q,0} \prod_{q=j}^{j+w-1} p_{c_i,q-j+1} \prod_{q=j+w}^{L} p_{c_i,q,0}$$

BG before motif · Motif · BG after motif

$$\rightarrow z_{ij} = \frac{P(X_i \mid Z_{ij}=1, \Theta^{t-1})}{\sum_{k=w+1}^{L} P(X_i \mid Z_{ik}=1, \Theta^{t-1})} \cdot \frac{P(Z_{ij}=1)}{P(Z_{ik}=1)}$$

→ Calculate Z Matrix: basically calculate all likelihoods and scale them to sum up to 1 for every sequence

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EM-Algorithm for Motif-discovery

Motif Position Z = Z_{ij} with Z_{ij} being probability that the motif starts at pos. j in seq. i

Motif Representation Θ = $p_{c,k}$, probability of character c at position k

Θ =

	0	1	2	3
A	0,25	0,2	0,3	0,1
C	0,24	0,3	0,5	0,2
G	0,23	0,1	0,1	0,4
T	0,25	0,4	0,1	0,3

BG | Motif

$$P(X_i \mid Z_{ij} = 1, \Theta) = \underbrace{\prod_{k=1}^{j-1} p_{c_i,k,0}}_{\text{Before Motif} = BG} \underbrace{\prod_{k=j}^{j+W-1} p_{c_i,k,1}}_{\text{Motif} = FG} \underbrace{\prod_{k=j+W}^{L} p_{c_i,k,0}}_{\text{After Motif} = BG}$$

→ Probability to find the Motif in sequence X_i at position j (condition $Z_{ij}=1$) for the given Motif model Θ.

Eg: 4A [GAA] ▽

Motif

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Updating the Model

→ count and weight the # of chars in the motif

↳ P of char c at pos k in motif

$$p_{c,k} = \frac{n_{c,k}}{\sum_b n_{b,k}} \rightarrow \#\ \text{Seq's}$$

$$n_{c,k} = \begin{cases} \sum_i \sum_{j \mid x_{ij}+k-1=c} z_{ij}^t, & k > 0 \\ n_c - \sum_{j=1}^{w} n_{c,j}, & k = 0 \end{cases}$$

→ Problems with EM:

→ once the probability is 0 you will never find that letter in the position of the motif → if motif not conserved of initial ones
↳ FAIL

→ trapped in local maxima

→ Every sequence has ~~to~~ have exactly one motif

→ can't find more than one motif

→ can't include previous knowledge

→ Extensions for MEME: inclusion of a pseudo count to avoid zero probabilities in the motif model

↳ starting position as similar as possible to actual motif

↳ ∀ distinct subseq of length W: → calculate the motif model
→ Run just one EM step

↳ Choose the model with the highest likelihood after one EM step as start model

↳ prior knowledge for good starting point (eg. A, T rich regions)

Gibbs Sampling

→ EM & MEME can be trapped in local maxima.

↳ P(local max) reduced by trying different starting points

→ Gibbs Sampling: random search, stochastic analog of EM

Difference: • In EM, we kept a distribution for starting points $\hat{z}_i$

• In Gibbs, we assign a specific starting point a_j to sequence X_i, but we will randomly resample these

Alg: I) Initial guesses of motif location

II) Score the model from those locations

III) Choose random sequence. Calculate likelihood of all possible motif starting positions

IV) Randomly choose a motif position from the probability distribution

V) Repeat 2-4 until convergence

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① Definitions

Sequence Motif:

→ a short, recurring pattern in DNA/RNA that is believed to have a biological function beyond that of a single sequence. These motifs are often involved in protein-DNA interactions that are critical to the regulation of gene expression.

Gene Knock-Out:

→ refers to the process of deactivating a gene by either replacing it with an inactive sequence or disrupting its function

→ advantages:

(+) allows study the function of a specific gene, understand the role of the gene in normal development and physiology as well as in the pathology of deseases.

(+) researches can gain insights into the biological processes that the gene is involved in.

Why would one want to knock out a gene encoding for a TF?

→ By knocking out a gene encoding for a TF, we can learn about the role that the factor plays in gene regulation and its effects on downstream biological processes

→ When we knockout a specific TF, we can see differentially expressed genes

→ Those genes are likely to be controlled by that specific TF and their upstream sequences are likely to contain the binding sites

(1)

b) Sequence Logo:

A sequence logo contains information about the frequency and conservation of each nucleotide / amino acid at each position within a sequence alignment. The height of each letter represents the frequency of that letter at that position.

c) Entropy distribution P_i vs relative entropy:

uncertainty (or entropy distribution P_i) measures the randomness in a distribution.

relative entropy measures the difference between two distribution.

② Expectation Maximization

Seq's : 1 C T G A A C G A G G
2 C A T T C C T T G G
3 C G G G C C G G A C
4 G T G G A C C C T C
5 A A A A T G T A C C

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a)

Probability Θ at any position in this genome to observe G / C

$$\Theta = \frac{total_count(C) + total_count(G)}{5 \cdot n} = 0,6$$

$$P(Seq_i \mid \Theta) = \Theta^{\#Head} \cdot (1-\Theta)^{\#Tail}$$

// do not take all orders

$P(Seq_1 \mid \Theta) = 0,6^6 \cdot 0,4^4 = 0,00119$ → likelihood to observe these seq's

$P(Seq_2 \mid \Theta) = 0,6^5 \cdot 0,4^5 = 0,00079$

$P(Seq_3 \mid \Theta) = 0,6^9 \cdot 0,4^1 = 0,00403$

$P(Seq_4 \mid \Theta) = 0,6^7 \cdot 0,4^3 = 0,00179$

$P(Seq_5 \mid \Theta) = 0,6^3 \cdot 0,4^7 = 0,00035$

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2)

Bayes = $P(Z_A|E) = \frac{P(E|Z_A)P(Z_A)}{P(E|Z_A)P(Z_A) + P(E|Z_B)P(Z_B)}$ if $P(Z_A) \neq P(Z_B)$ → $= \frac{P(E|Z_A)}{P(E|Z_A)+P(E|Z_B)}$

3) $\Theta_A^0 = 0,4$; $\Theta_B = 0,6$

$P(A|A) = B(\Theta, n, k)$, $P(A|A)$ is calculated using a method/table from [Gr, C]

First (C, G) → H & (A, T) → T

→ Iteration 1:

1) Calculate $P(Seq_i|\Theta_x^0)$ with binomial distribution.

		$P(Seq_i\|\Theta_A^0)$	$P(Seq_i\|\Theta_B^0)$
Seq 1:	HTHTTHHTHH	0,1111	0,25
Seq 2:	HTTTHHTTHH	0,2	0,2
Seq 3:	HHHHHHHHTH	0,0016	0,04
Seq 4:	HTHHTHHHTH	0,0422	0,215
Seq 5:	TTTTTHTTHH	0,2765	0,0422

Both coil types appear equally frequently

2) $P(\Theta_A|Seq_i) = \frac{\Theta_A^h (1-\Theta_A)^t}{\Theta_A^h (1-\Theta_A)^t + \Theta_B^h (1-\Theta_B)^t} = \frac{P(Seq_i|\Theta_A)}{P(Seq_i|\Theta_A) + P(Seq_i|\Theta_B)}$

	$P(\Theta_A\|Seq_i)$	$P(\Theta_B\|Seq_i)$	#Heads for A	#Heads for B
1	0,307	0,693	1,8	3,158
2	0,5	0,5	2,5	2,5
3	0,0243	0,976	0,218	8,782
4	0,1634	0,837	1,144	5,8562
5	0,837	0,1634	5,8562	1,144

$\Theta_A^1 = \frac{\sum \#\text{Head attributed to A}}{n * \sum P(\Theta_A^0|Seq_i)} = 0,674$

$\Theta_B^1 = \frac{\sum \#\text{Head attributed to B}}{n * \sum P(\Theta_B^1|Seq_i)} = 0,303$

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c) EM used for: → motif discovery
→ discover CpG-islands

d) Problems with EM

(-) If/once the letter probability of a certain letter equals 0, we will never find a motif with that letter at that position
↳ if motif is not a combination of the Iterable queries → FAIL

(-) can be trapped in local maxima

(-) there has to be exactly one motif in every sequence

(-) can not find more than one motif

(-) can not include prior knowledge

↳ MEME: → introduces pseudocounts to prevent zeros
→ can include prior knowledge ⇒ good starting point
↳ But can be trapped also in local maxima